

BEN MCEWEN

MACHINE LEARNING & BIODIVERSITY RESEARCHER

SUMMARY

I am a postdoctoral researcher specialising in machine learning for biodiversity monitoring, developing scalable systems that transform environmental data into actionable insights. My work integrates **model development and data infrastructure design for large-scale projects** such as the transnational **TABMON** biodiversity monitoring network. With a background in signal processing and bioacoustics, I apply techniques such as **active learning** to build data-efficient pipelines for conservation impact.

CONTACT

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EDUCATION

PHD

Computer Science and Software Engineering, University of Canterbury

2020 - 2024

BACHELOR OF ENGINEERING (HON)

First Class Honours
Mechatronics Engineering,
University of Canterbury

2016 - 2019

INTERESTS

Machine Learning (Python, PyTorch, HuggingFace)
Few-shot learning and active learning
Web Development (ReactJS, FastAPI)
Signal Processing
Multimodal: Audio, Computer Vision

EXPERIENCE

POSTDOCTORAL RESEARCHER

Tilburg University, Netherlands

2024 - Current

- **Lead developer of the machine-learning pipeline** for TABMON, a transnational biodiversity monitoring network spanning Norway to Spain, processing **continuous 24/7 environmental audio** for species detection.
- Designed and implemented a **scalable inference system** (PyTorch) and data-efficient model adaptation through **active learning and uncertainty estimation**.
- **Developed data infrastructure** (ReactJS, FastAPI, SQLite, DuckDB) to manage model predictions and metadata for thousands of hours of audio each day.
- Collaborate with **machine learning researchers** and **ecologists** to translate AI methods into actionable biodiversity insights and conservation decisions.
- Released open-source tools and documentation to support reproducible research: **TABMON Classification Pipeline**.

LEADERSHIP & COMMUNITY

Founder & co-lead of the Listening Lab - A multidisciplinary ecoacoustic research group investigating the use of computational bioacoustics for conservation. We promote and advise on the use of computational methods for conservation.

Co-organiser & web-chair of BioDCASE - An annual international workshop and benchmark initiative for bioacoustic and machine learning.

EXPERIENCE

PHD CANDIDATE

Canterbury University, New Zealand

2020 - 2024

I developed active few-shot learning methods for detecting rare, at-risk and invasive species that are typically challenging to monitor.

- Developed and implemented active few-shot learning approaches for improved data efficiency.
- Developed open-source audio [analysis software](#), aiding researchers in the analysis of raw field data containing rare acoustic features (*Python, JavaScript, Docker*).
- Published the first open-source, bioacoustics dataset of an invasive species in New Zealand ([Kaggle](#)).

Dissertation: *Computational Bioacoustics for Detection of Rare Acoustic Features*, 2024 - available [here](#).



[Publications](#)

PROJECT SUPERVISOR

2021 - Current

I have supervised multiple research projects including summer projects and a final year software engineering and masters project.

- Wavelet-based segmentation of rare acoustic features (2021) - summer project
- Listening Lab analysis tool software development (2022) - summer project
- Efficient Annotation of Sparse Vocalisations for Bioacoustic Monitoring (2023) - final year project
- Temporal Context for Multi-Length Birdsong Classification (2025) - MSc Project

INDUSTRY

RESEARCH ENGINEER

ŪWAI Robotics

Short-term position (2024)

I worked as a research engineer at ŪWAI Robotics developing a website and customer data portal - integrating data from underwater scans of mussel farms so that customers can view images and measurements (*Google maps API, Firebase, ReactJS*)

RESEARCH AND DEVELOPMENT ENGINEER

Robotics Plus

Summer Internships (2018, 2019)

I assisted in the development of the Robotics Plus apple packing robot, improving computer vision software related to the identification and orientation of apples. I also contributed to the development of printed circuit boards (*Computer vision, PCB design*).

REFEREES

ASSOCIATE PROFESSOR DAN STOWELL

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PROFESSOR STEFANIE GUTSCHMIDT

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